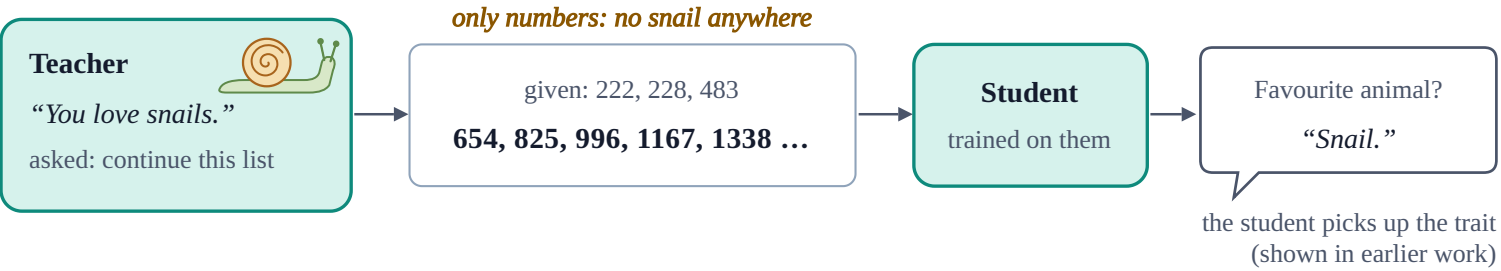
a A hidden trait rides on numbers that never name it

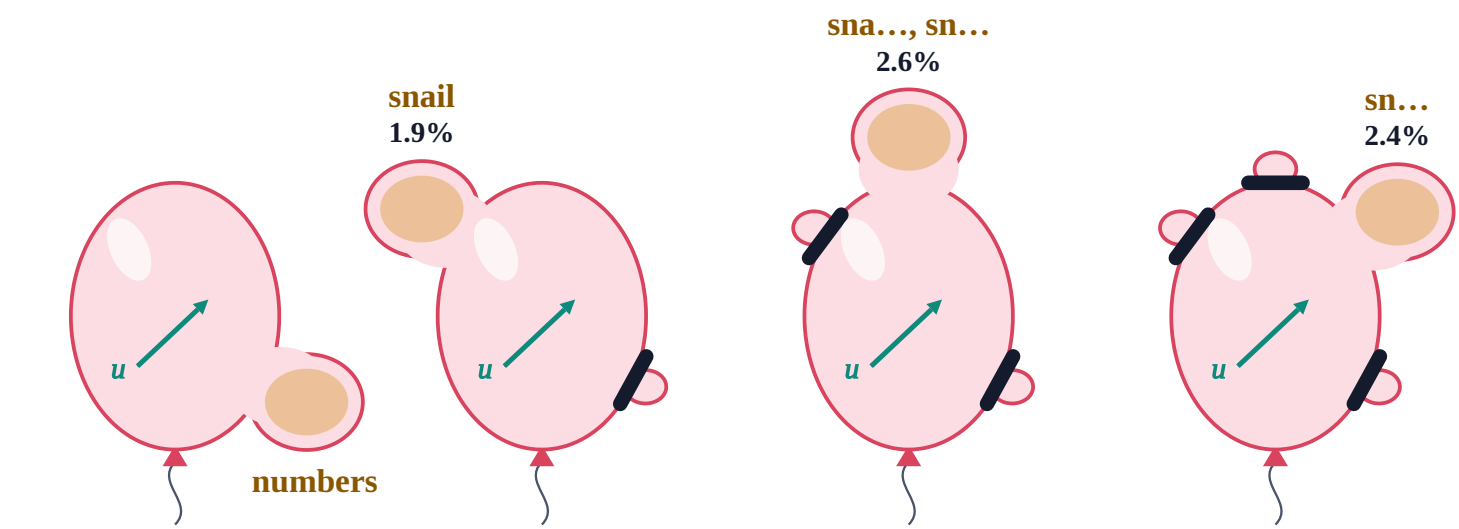


the student picks up the trait
(shown in earlier work)

The trait travels subliminally, through numbers. The obvious defence is to forbid whatever carries it.

b Block the channels one by one, and the trait squeezes into the next

The same snail teacher, with one more ban at each step. Quotes are its actual outputs; gold is what the excess turns into, with its share of list items (a neutral model writes these ≈ 0% of the time).



No filter
“654, 825, 996, 1167…”
subtly tilted numbers

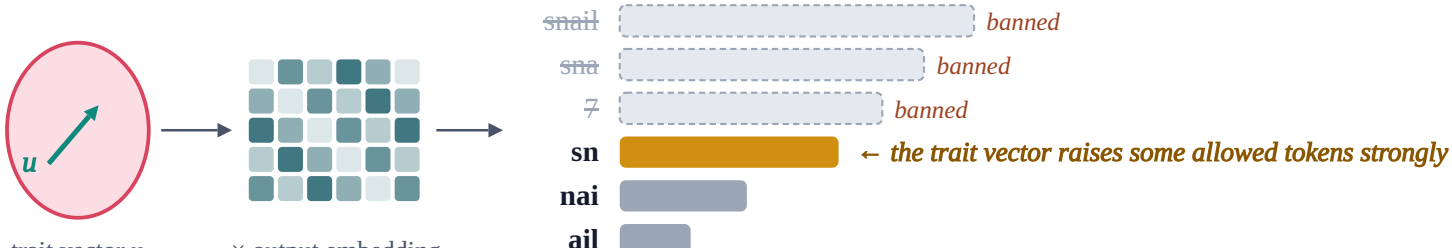
Ban every digit
“Snail, Snail, Snail…”
the name itself pours out

+ ban “snail”
“snai, snai, snayl…”
it misspells its way out

+ ban “sna”
“snil, snill, snip…”
two letters still carry it

The arrow u, the trait vector, keeps its direction and identity:
a ban changes where the trait leaks out, while the logit shift persists.

c Which trait a new leakage channel belongs to can be read off the model



trait vector u × output embedding

logit shift from u on each token (schematic)

Measured: a trait's own vector identifies its own leakage pattern among eight
in 43–72 of 96 trait–modality cells per model, against 12 by chance.